

St. Margaret's Co-educational English Secondary and Primary School

S.3 Information and Communication Technology

Mock Practice Paper · Final Written Examination & Quiz

Name: _____

Date: _____

Class: S.3____ Class No.: _____

Time Allowed: 40 minutes

Total: 50 marks

Instructions:

1. Answer ALL questions in this Question-Answer Paper.
2. This paper must be answered in English.
3. For multiple-choice questions, circle the BEST answer only.
4. For fill-in-the-blank questions, write the exact term on the line provided.
5. For short-answer and extended questions, answer in full sentences where appropriate.
6. Write legibly.

Quiz

Part A: Matching (5 marks)

Match each Python code snippet on the left to its correct description on the right. Write the letter of the correct description in the Answer column.

Descriptions:

- A. Assigns a float (decimal) value to a variable.
- B. Performs subtraction and stores the result in a variable.
- C. Captures user input and assigns it to a variable as a string.
- D. Outputs a literal string or variable to the console.
- E. Converts a value into an integer number.

#	Code Snippet	Answer (write letter)
1.	<code>total = a - b</code>	
2.	<code>ans = input("?")</code>	
3.	<code>price = 19.99</code>	
4.	<code>print("Hello")</code>	
5.	<code>int(n)</code>	

Quiz — Part B: Short Questions (5 marks)

B1. In MIT App Inventor's personal image classifier tool, a model outputs "Carrot Cake 0.97363" and "Siu Mai 0.02603". Which item will the machine classify the image as, and why? (2 marks)

B2. When training an image classifier model, a student uploads only 3 images per label. Suggest **two** ways to improve the model's accuracy. (2 marks)

B3. Name the **non-visible** App Inventor component used to announce classification results out loud. (1 mark)

Written Exam

Part A: Multiple-Choice Questions (10 marks, 1 mark each)

Answer all questions. Choose and *CIRCLE* the best option for each question.

1. Which of the following is a valid Python variable name?
 - A. 3score
 - B. my-score
 - C. _total
 - D. my score
2. A student writes: `print("I study S" + 3)`. What error will occur?
 - A. SyntaxError
 - B. TypeError
 - C. ValueError
 - D. No error — it prints "I study S3"
3. Which data type does the `input()` function always return in Python?
 - A. Integer (int)
 - B. Float (float)
 - C. Boolean (bool)
 - D. String (str)
4. Which of the following describes **Knowledge-based Reasoning** in AI robots?
 - A. Fast, direct actions based on instinctive reactions without conscious thought.
 - B. Actions derived from instructions or patterns learned from prior experience.
 - C. Actions from in-depth analysis using prior experience, process knowledge, and objectives.
 - D. Actions based solely on random trial-and-error experimentation.
5. According to the World Economic Forum's "Future of Jobs Report 2020", which of the following jobs is expected to have **decreasing** demand by 2025?
 - A. AI and Machine Learning Specialists
 - B. Big Data Specialists
 - C. Data Entry Clerks
 - D. Internet of Things Specialists
6. Strings in Python are described as "immutable". What does this mean?
 - A. Strings can be changed at any time by reassignment.
 - B. Once declared, individual characters in a string cannot be modified directly.
 - C. Strings always convert to integers automatically.
 - D. Strings can only contain numbers.
7. In the App Inventor personal image classifier, what is the purpose of the **WebViewer** component?
 - A. To store training images in the cloud.
 - B. To display the real-time camera feed so users can aim their phone at objects.

- C. To announce classification results out loud.
 - D. To download the trained model from the server.
8. What is the term for the gap between those with ICT access and those without?
- A. Netiquette
 - B. Repetitive Strain Injury
 - C. The Global Digital Divide
 - D. Text Neck
9. In supervised learning (SL), the AI model learns from:
- A. Trial and error without any labelled data.
 - B. Paired input and output training data provided by a supervisor.
 - C. Observing the environment and receiving random rewards.
 - D. Only the output data, without any input examples.
10. Which physical health problem is caused by damage to nerves and tendons from prolonged ICT use, such as Carpal Tunnel Syndrome?
- A. Text Neck
 - B. Eye Strain
 - C. Repetitive Strain Injury (RSI)
 - D. Global Digital Divide

[illegible]

Part B: Matching (10 marks, 1 mark each)

Match each Python code snippet to its correct description. Write the letter in the Answer column.

Descriptions:

- A. A conditional check for less than or equal to.
- B. Assigns an integer value to a variable.
- C. Checks the data type of a variable.
- D. Assigns a string (text) value to a variable.
- E. A conditional check to see if x is not equal to 5.
- F. Performs subtraction and stores the result.
- G. Assigns a float value to a variable.
- H. Outputs a literal string to the console.
- I. Captures user input as a string.
- J. Converts a value into an integer.

#	Code Snippet	Answer (write letter)
1.	<code>score = 100</code>	
2.	<code>city = "London"</code>	
3.	<code>type(x)</code>	
4.	<code>if not x == 5:</code>	
5.	<code>if x <= y:</code>	
6.	<code>price = 19.99</code>	
7.	<code>total = a - b</code>	
8.	<code>print("Hello")</code>	
9.	<code>ans = input("?")</code>	
10.	<code>int(n)</code>	

Part C: Fill in the Blanks (10 marks, 1 mark each)

Write the correct word or phrase on each line provided.

C1. In machine learning, the process of allowing the model to learn from provided images is called _____.

C2. The extension file that must be imported into App Inventor so that the trained model can be used is called _____.

C3. Two main approaches for training AI models with simulations are _____ Learning and _____ Learning.

C4. AI robots primarily make decisions based on _____-based Reasoning.

- C5. Python uses _____ to determine which groups of statements belong together in a code block.
- C6. The physical health problem caused by pain from improper head positioning while using mobile devices is called _____.
- C7. Attempting to concatenate a string with an integer using + in Python will result in a _____ error.
- C8. The OECD "Future of Education and Skills 2030" project proposes three classes of skills: Social and Emotional Skills, _____ Skills, and Practical and Physical Skills.
- C9. Equity of Access refers to ensuring everyone has equal opportunity to benefit from ICT, regardless of their _____.
- C10. A variable in Python must begin with a _____ or an underscore, not a number.

Part D: Questions (20 marks)

D1. App Development with MIT App Inventor (6 marks)

- (a) A student wants to build a food recognition app using the personal image classifier tool. List the **six** steps to create the training model before integrating it into App Inventor. (3 marks)

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- (b) State **two** required components specific to the personal image classifier App Inventor project (not Buttons or Labels). (2 marks)

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- (c) A student trains a model with only photos of dim sum taken from directly above, all under identical bright lighting. The model works well in testing but fails in real use. Suggest **one** reason and how to fix it. (1 mark)
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D2. Python Basics

(8 marks)

(a) Study the following code. Identify **two** errors and write the corrected version.

(4 marks)

```
1 A = input()
2 B = 3 * A
3 print("The answer is: " + B)
```

Error 1:

Error 2:

Corrected code:

(b) Write a Python program that asks the user for their name and their class number, then prints:

Hello [name], you are in class S[number].

(3 marks)

(c) State the data type returned by each expression and give a reason.

(1 mark)

(i) ``type(True)`` → _____ (ii) ``type(3.0)`` → _____

D3. AI and the Future of Work

(6 marks)

- (a) Describe how AI is changing the future of work. In your answer, mention **both** a negative and a positive impact. (2 marks)
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- (b) The three levels of reasoning in AI robotic systems are Skill-based, Rule-based, and Knowledge-based. A food delivery robot is tasked with delivering to four houses in the most efficient route. Explain which reasoning level is **most suitable** and why. (2 marks)
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- (c) According to the OECD "Future of Education and Skills 2030", identify **two** skills from **different** skill categories that students should develop to thrive in a future with AI. Explain why each skill is important. (2 marks)
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End of Paper